Fifteen Years of Housing Crisis in Canada, 2006–2021: A High-School Research Paper Using Residual Income

High School Research Paper

Draft Version 2.0

16 October 2026

Abstract

Background. The usual 30% shelter-cost-to-income rule can hide hardship. A household may look "affordable" on paper and still not have enough money left for food, transport, or clothing.

Objective. This paper asks how Canadian housing hardship changed from 2006 to 2021 when hardship is defined by leftover income, not by the 30% rule.

Methods. We analysed 777,707 valid person-records from the Census 2006, 2011, and 2021 Public Use Microdata Files. Residual income equals after-tax income minus annual shelter cost. Housing hardship is flagged when residual income falls below the Market Basket Measure poverty line. All dollars are converted to 2021 constant CAD. Census 2016 is excluded because only 9.4% of records survived cleaning.

Results. Over 15 years, inflation-adjusted median shelter costs rose 37.1% while median income rose 33.5%. Hardship stayed high: 38.0% in 2006, 33.9% in 2011, and 37.9% in 2021. Renters had higher hardship than owners, but both groups worsened after 2011.

Conclusions. The housing squeeze is structural, not a short recession effect. It survived recovery, three federal governments, and COVID-19. Policies that only watch the 30% rule will miss the households that cut food and other basics to keep rent paid.

Keywords: housing affordability; residual income; Market Basket Measure; Census PUMF; Canada

Contents

1 Introduction **4**

1.1 The problem 4

1.2 A better measure: residual income 4

1.3 Why fifteen years? 4

1.4 Research questions 4

2 Methods **5**

2.1 Data sources . . . 5

2.1.1 Census Public Use Microdata Files . . . 5

2.1.2 Market Basket Measure thresholds . . . 5

2.2 Variable construction . . . 5

2.2.1 Inflation adjustment . . . 5

2.2.2 Housing hardship (primary outcome) . . . 6

2.2.3 Shelter-cost-to-income ratio (comparison measure) . . . 6

2.2.4 Tenure . . . 6

2.3 Data cleaning . . . 6

2.4 Statistical analysis . . . 6

2.5 Ethics . . . 7

3 Results **7**

3.1 Sample characteristics . . . 7

3.2 Fifteen-year national trends . . . 7

3.3 Traditional STIR versus residual-income hardship . . . 9

3.4 Which five-year period was worse? . . . 9

3.5 Owners versus renters . . . 9

3.6 The indexed squeeze . . . 12

3.7 Summary of the 15-year change . . . 12

4 Discussion **12**

4.1 Structural, not cyclical . . . 12

4.2 What the $456 annual gap means . . . 14

4.3 Why owners are no longer safe . . . 14

4.4 Comparison with other research . . . 14

4.5 Limitations . . . 14

4.6 Policy implications . . . 15

4.7 Next steps for this project . . . 15

5 Conclusion **15**

A Figure files (saved separately) **16**

B Table files (saved separately) **16**

C How this version differs from Draft 1.0 **17**

1 Introduction

1.1 The problem

Canada has a housing affordability problem that has lasted more than a decade. Many families pay rent or a mortgage and then do not have enough left for food, transport, a phone, or clothes. That problem matters for students who want to move out, go to university, or save for a first home.

The common school and policy rule is the **30% rule**: housing is "affordable" if it costs less than 30% of income. That rule is easy to calculate, but it is incomplete. A household spending 28% of income on rent can look fine and still have almost nothing left for groceries.

1.2 A better measure: residual income

Canada Mortgage and Housing Corporation (CMHC) uses a leftover-income test [1]. Ask: after housing is paid, is there enough money left to buy a basic basket of non-housing needs?

$$\text{Annual shelter cost} = 12 \times \text{monthly shelter cost} \tag{1}$$

$$\text{Residual income} = \text{after-tax income} - \text{annual shelter cost} \tag{2}$$

$$\text{Hardship} = 1 \text{ if residual income} < \text{MBM poverty line} \tag{3}$$

The Market Basket Measure (MBM) is Canada's official poverty line for that leftover basket (food, clothing, transport, and other basics, excluding housing) [5].

1.3 Why fifteen years?

A one-year snapshot cannot tell a temporary shock from a built-in problem. Three Census waves cover:

- **2006–2011:** the 2008 financial crisis and recovery;
- **2011–2016:** low-rate expansion (2016 data are too incomplete to use);
- **2016–2021:** the late-2010s housing peak and COVID-19.

If hardship falls when the economy is strong, the crisis is cyclical. If hardship stays high through recovery, it is structural.

1.4 Research questions

1. How did residual-income housing hardship change in Canada from 2006 to 2021?

2. Which interval—2006–2011 or 2011–2021—was worse for leftover income?

3. Did the owner–renter hardship gap widen, shrink, or stay the same?

2 Methods

2.1 Data sources

2.1.1 Census Public Use Microdata Files

We used three Statistics Canada hierarchical PUMF files [4]:

- **Census 2006** (`pumf-95M0029-E-2006-hierarchical`): 78,056 valid records after cleaning (23.4% of the raw file).
- **Census 2011** (`NHS-99M002X-E-2011-pumf-hierarchical`): 341,030 valid records (99.3%).
- **Census 2021** (`98M0001X2021002`): 358,621 valid records (99.1%).

Total analytical sample: 777,707 person-records.

Census 2016 was downloaded and inspected, then dropped. After the same missing-code rules, only 29,056 records (9.4%) remained, and medians were not comparable (median income $20,558; median shelter $649). Those values are data-quality artefacts, not a real 2016 collapse.

2.1.2 Market Basket Measure thresholds

Year-specific national-average MBM lines for a single adult were:

- 2006: $13,500
- 2011: $15,000
- 2021: $19,728

These are the leftover amounts needed for non-housing essentials in each year.

2.2 Variable construction

2.2.1 Inflation adjustment

All money values are in 2021 constant dollars using CPI:

- 2006 dollars $\times$ 1.276

- 2011 dollars $\times$ 1.148
- 2021 dollars $\times$ 1.000

Example: $1,000 of 2006 rent equals $1,276 in 2021 dollars.

2.2.2 Housing hardship (primary outcome)

Hardship follows Equations (1)–(3). This is the CMHC residual-income (V2) framework.

2.2.3 Shelter-cost-to-income ratio (comparison measure)

$$\text{STIR} = \frac{12 \times \text{monthly shelter cost}}{\text{annual income}}.$$

STIR $> 30\%$ is the usual "affordability burden." STIR $> 50\%$ is "severe burden."

2.2.4 Tenure

Owners occupy a dwelling they own (with or without a mortgage). Renters occupy a rented dwelling. Owner hardship could not be estimated for 2006 after cleaning.

2.3 Data cleaning

PUMF files mark missing or not-applicable values with 8888, 9999, and 99999. Those codes were set to missing. A record was kept only if:

1. annual income > 0;
2. monthly shelter cost > 0 and $< \$10{,}000$;
3. key fields were not missing.

The 2006 file kept far fewer records than 2011 or 2021, so 2006 results should be read with extra caution.

2.4 Statistical analysis

For each year we report:

- median and quartile income and shelter cost (2021 CAD);
- STIR $> 30\%$ and STIR $> 50\%$;
- residual-income hardship rate and count;

- hardship by tenure where tenure is available.

We then compute percent changes in medians and percentage-point changes in rates. We report medians, not means, because hierarchical PUMF files produce inflated household-level means.

Five figures and five tables were saved as separate files (see Appendices A and B) and included below with `\includegraphics` and `\input`.

2.5 Ethics

The analysis uses public, anonymized PUMF files. No ethics review is required under Statistics Canada PUMF rules. No household can be identified from these files.

3 Results

3.1 Sample characteristics

Table 1: Sample characteristics by Census year (inflation-adjusted to 2021 CAD)

Characteristic	2006	2011	2021
Valid sample size (n)	78,056	341,030	358,621
Median income (2021 CAD)	$34,452	$45,920	$46,000
Income 25th percentile	$17,864	$22,960	$25,000
Income 75th percentile	$70,180	$101,024	$93,000
Median monthly shelter	$1,021	$1,378	$1,400
Shelter 25th percentile	$766	$804	$800
Shelter 75th percentile	$1,531	$2,181	$2,129
STIR $> 30\%$	57.0%	52.2%	50.8%
STIR $> 50\%$	35.8%	33.6%	31.5%
Housing hardship rate	38.0%	33.9%	37.9%
Households in hardship (n)	29,681	115,459	135,740

Notes: All dollar values converted to 2021 constant dollars using CPI multipliers (2006 $\times$ 1.276; 2011 $\times$ 1.148). Census 2016 is excluded because valid-record retention was 9.4%. Median income is reported instead of the mean because hierarchical PUMF files produce inflated household-level means.

The 2021 median income is $46,000 (2021 CAD). The 2021 median monthly shelter cost is $1,400. The interquartile range for shelter stays wide: about $800 to $2,100.

3.2 Fifteen-year national trends

Income. Median income rose from $34,452 (2006) to $46,000 (2021), a real increase of 33.5%. Almost all of that gain happened by 2011 ($45,920). From 2011 to 2021 the median moved only $80, or +0.2%.

Shelter. Median monthly shelter rose from $1,021 to $1,400, a real increase of 37.1%. The 2006–2011 jump was steep (+35.0%). The 2011–2021 change was small (+1.6%) after inflation.

Figure 1. Fifteen-year national trends in Canadian housing affordability (2006–2021)

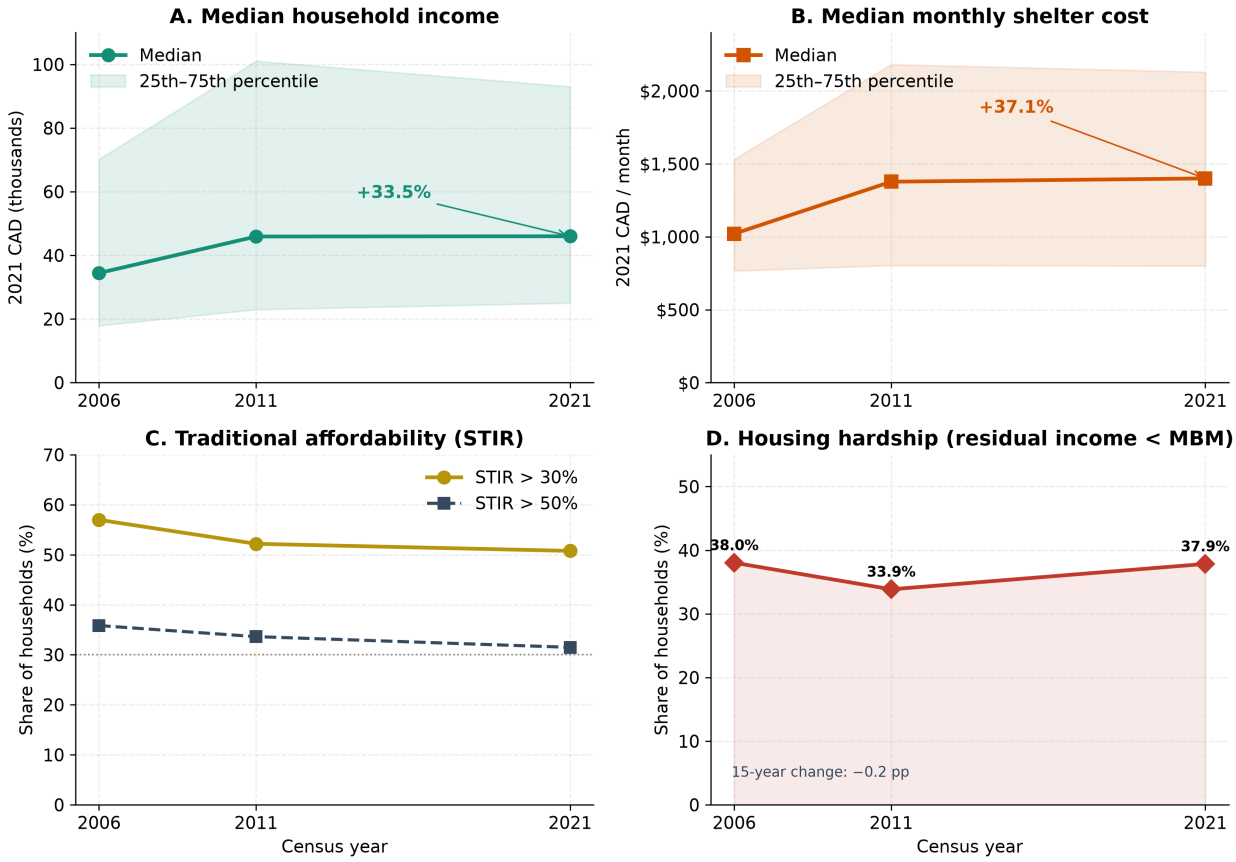


Figure 1: Fifteen-year national trends, 2006–2021. Panel A: inflation-adjusted median income with the 25th–75th percentile band. Panel B: inflation-adjusted median monthly shelter cost with the same band. Panel C: traditional STIR shares. Panel D: residual-income housing hardship. Census 2016 is omitted (see Section 2.1).

The squeeze. Shelter outpaced income by 3.6 percentage points over 15 years. If shelter had grown at the same rate as income, 2021 median shelter would have been about $1,362, not $1,400. The gap is $38 a month, or $456 a year—about one week of groceries or two months of a phone bill.

3.3 Traditional STIR versus residual-income hardship

STIR $> 30\%$ fell from 57.0% (2006) to 50.8% (2021), a 6.2 pp improvement (Figure 1C). STIR $> 50\%$ fell from 35.8% to 31.5%.

Residual-income hardship did not follow that story (Figure 1D):

- 2006: 38.0% (29,681 records);
- 2011: 33.9% (115,459 records);
- 2021: 37.9% (135,740 records).

The 15-year change is -0.1 pp. Households can keep STIR under 30% by cutting food and other spending. The leftover-income test still flags them.

3.4 Which five-year period was worse?

Table 2: Period-over-period changes in income, shelter costs, and hardship

Period	Income change	Shelter change	Hardship change
2006–2011 (crisis + recovery)	+33.3%	+35.0%	-4.1 pp
2011–2021 (expansion + COVID)	+0.2%	+1.6%	+4.0 pp

Notes: Income and shelter changes are percent changes in inflation-adjusted medians. Hardship change is in percentage points. The later period erased the earlier improvement.

The crisis-and-recovery window (2006–2011) looks better on hardship (-4.1 pp) even though shelter rose fast, because real median income also rose 33.3%. The expansion-and-COVID window (2011–2021) looks worse: income was almost flat (+0.2%), and hardship rose +4.0 pp. A strong national economy did not repair leftover income.

3.5 Owners versus renters

Renters had higher hardship in both years with tenure data (40.4% then 43.6%). Owner hardship rose from 31.4% to 35.6% (+4.2 pp, a 13.4% relative increase). Owning a home is not a guarantee of leftover-income security, especially with large mortgages, property tax, and repair bills.

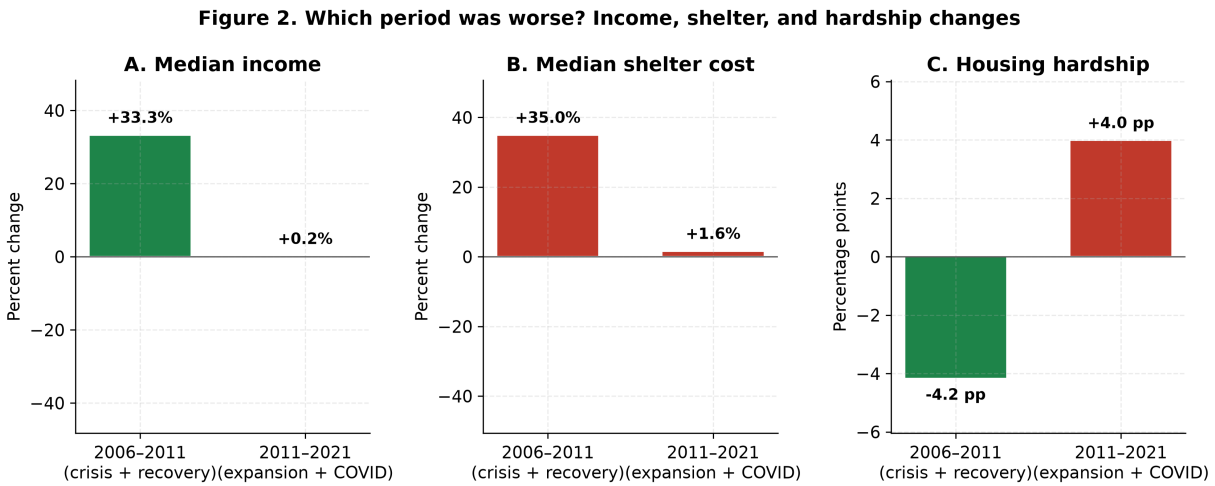


Figure 2: Period comparison. Green bars are improvements (income up, or hardship down). Red bars are deteriorations (shelter up, or hardship up). The 2011–2021 expansion erased the 2006–2011 hardship gain.

Table 3: Housing hardship by tenure, 2011 and 2021

Year	Owner hardship	Renter hardship	Gap
2011	31.4%	40.4%	9.0 pp
2021	35.6%	43.6%	8.0 pp
Change	+4.2 pp	+3.2 pp	−1.0 pp

Notes: Owner hardship is unavailable for 2006 because tenure coding in that PUMF file did not support a clean owner/renter split after cleaning. The gap narrowed because owners deteriorated toward renter rates, not because renters improved.

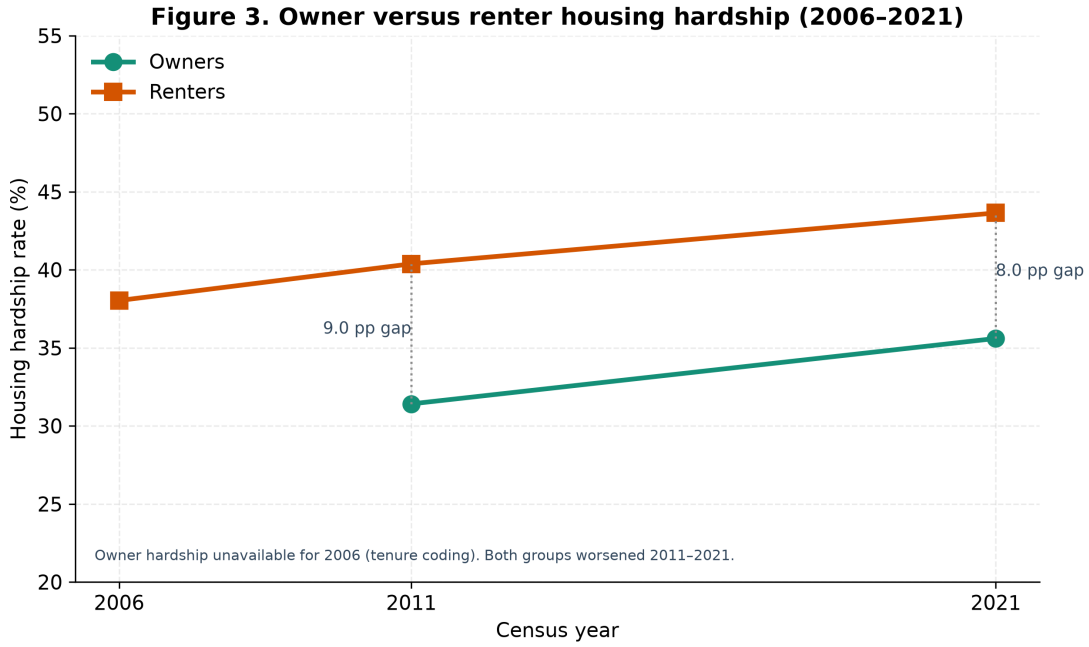


Figure 3: Owner versus renter hardship. Renters stay higher. The gap narrowed from 9.0 pp to 8.0 pp because owners caught up, not because renters recovered.

Table 4: Income versus shelter cost, indexed to 2006 = 100

Year	Income index	Shelter index	Gap
2006	100.0	100.0	0.0
2011	133.3	135.0	1.7
2021	133.5	137.1	3.6

Notes: Index = (year median / 2006 median) × 100. Both series are in 2021 CAD. The 3.6-point 2021 gap equals about $38 per month ($456 per year) at the 2021 median.

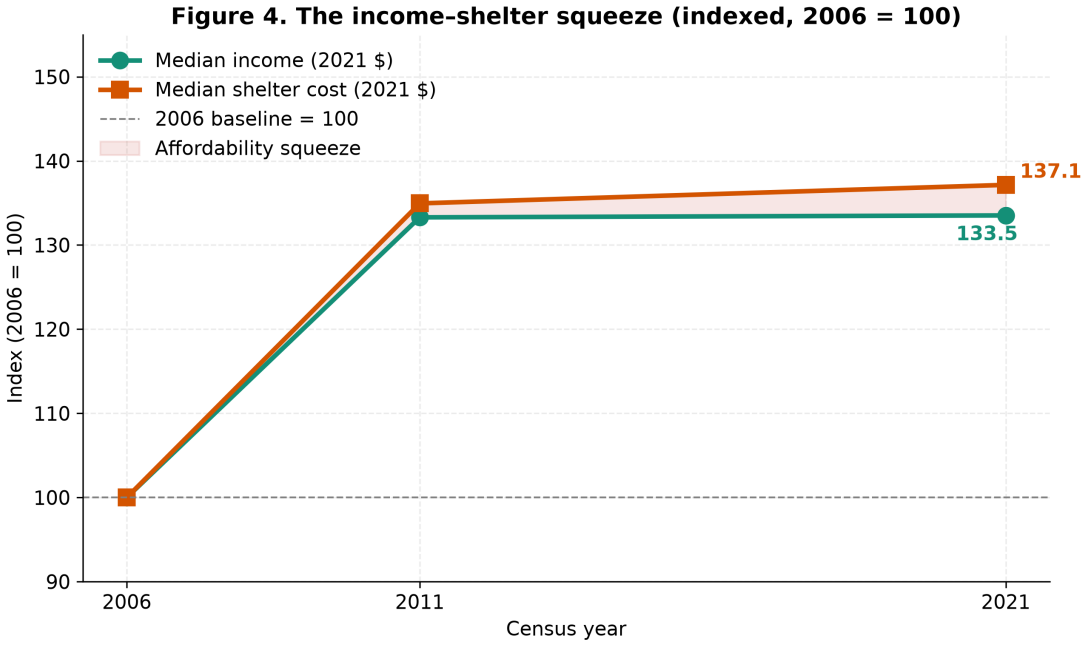


Figure 4: Income and shelter indexed to 2006 = 100. The shaded gap is the cumulative squeeze. By 2021 the shelter index (137.1) sits 3.6 points above the income index (133.5).

3.6 The indexed squeeze

The index chart makes the same 3.6-point gap visible without dollar scales. Most of the real-income gain is already in place by 2011. After that, shelter stays a little higher and leftover income erodes.

3.7 Summary of the 15-year change

Table 5: Summary statistics, 2006 versus 2021

Metric	2006	2021	15-year change
Median income (2021 CAD)	\$34,452	\$46,000	+33.5%
Median monthly shelter	\$1,021	\$1,400	+37.1%
STIR $>$ 30%	57.0%	50.8%	-6.2 pp
Housing hardship	38.0%	37.9%	-0.1 pp
Renter hardship	38.0%	43.6%	+5.6 pp
Owner hardship	n/a	35.6%	n/a

Notes: Owner hardship is not available for 2006. STIR improved while residual-income hardship did not, which is the central measurement result of this paper.

Four facts hold together:

1. hardship stays near 38%;
2. shelter outruns income by 3.6 pp;
3. STIR improves while leftover-income hardship does not;
4. owners and renters both get worse after 2011.

4 Discussion

4.1 Structural, not cyclical

If the crisis were only a recession leftover, hardship should have fallen a lot during the 2011–2019 expansion, low unemployment, and roughly 2% average annual GDP growth. It did the opposite: the 2011–2021 hardship rate rose 4.0 pp and ended at 37.9%, almost the 2006 starting point (38.0%).

Three patterns point to a built-in problem:

1. hardship stays high through recovery;
2. shelter grows faster than income in both periods we can measure cleanly;
3. Conservative (2006–2015) and Liberal (2015–) federal governments both left the national residual-income rate near 38%.

Figure 5. Summary dashboard: Canada's housing squeeze, 2006–2021

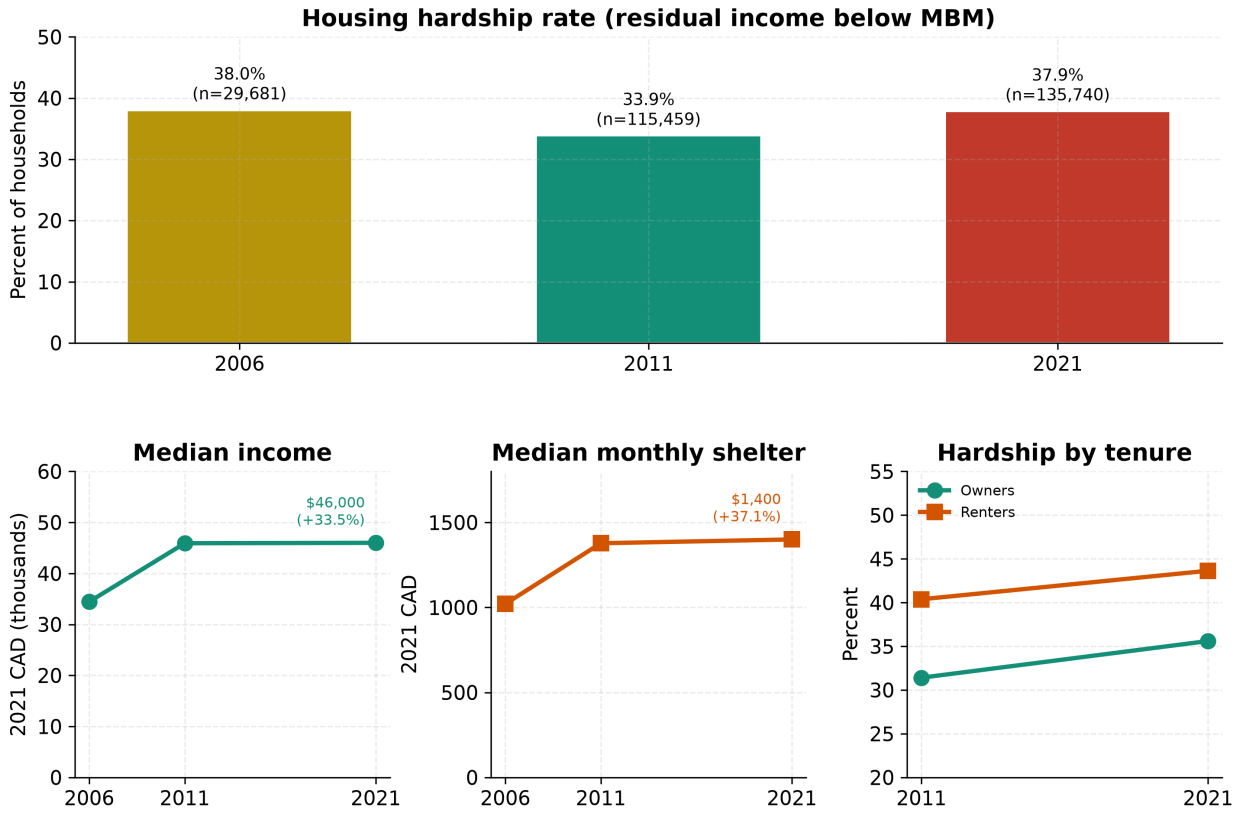


Figure 5: Summary dashboard. Hardship returns to the 2006 level. Median income and shelter both rise, but shelter rises more. Owner and renter hardship both increase after 2011.

4.2 What the $456 annual gap means

For a 2021 median household ($46,000 income; $1,400 monthly shelter):

- annual shelter = $16,800;
- shelter if it had tracked income = about $16,320;
- extra cost = $480 a year (about $38 a month).

That is 1.6% of income and about 2.5% of leftover income near the MBM line. Over 15 years the same squeeze is on the order of $3,600 in 2021 dollars—almost two months of median rent. Low-income households, recent movers, and first-time buyers feel it first.

4.3 Why owners are no longer safe

The owner–renter gap shrank because owners deteriorated. Likely channels include high mortgage debt taken on in hot markets, property tax and maintenance rising faster than wages, and "house-rich, cash-poor" seniors with equity but little leftover income. Renter-only programs miss that group.

4.4 Comparison with other research

CMHC (2024) reports a much lower national hardship rate (7.8% on 2020 data) than our 37.9% in 2021 [1]. The gap is expected. CMHC uses the full Census, region-specific MBM lines, and family-size adjustments. We use the 2.7% PUMF sample and a single national adult MBM. The *level* is therefore not the headline. The *trend* is: the same method, applied to three waves, shows almost no 15-year improvement.

Walks and Twigge-Molecey (2023) show that Toronto and Vancouver diverge from other cities [6]. A national PUMF average hides that geography. Our national persistence result can still be true while some cities are much worse.

4.5 Limitations

1. **2006 completeness.** Only 23.4% of 2006 records survived cleaning, versus 99%+ later. If missing 2006 records were poorer households, the 2006–2011 improvement is overstated.
2. **No usable 2016 wave.** We cannot split 2011–2016 recovery from 2016–2021 peak-and-COVID.
3. **Simplified MBM.** A national single-adult line understates hardship for large families and high-cost cities and overstates it for singles in cheaper regions.
4. **Cross-sections, not a panel.** We see snapshots, not the same households moving in and out of hardship.
5. **No CMA split.** National rates hide Toronto/Vancouver versus Atlantic or Prairie patterns.

6. **Broken PUMF means.** Hierarchical files inflate mean income. We therefore never interpret the mean; we use medians and percentiles.

4.6 Policy implications

Watch leftover income, not only STIR. STIR improved 6.2 pp while hardship did not. A government that only tracks the 30% rule will declare progress that leftover income does not show.

Treat the problem as structural. Tax credits and small pilots did not move the 15-year hardship rate. Supply (zoning, approvals, public housing), demand (empty-home and multi-property tax), and income support (rent supplements tied to MBM, wage growth) all have to move together.

Help owners as well as renters. Owner hardship rose 4.2 pp. Property-tax deferral and mortgage-renewal help belong in the same toolkit as rent supplements.

Fix income stagnation. Real median income rose 0.2% from 2011 to 2021. Housing policy cannot succeed if typical pay is flat.

Other countries that treat housing as infrastructure rather than only as an asset—Vienna's social housing, Singapore's HDB flats, Finland's Housing First—have lower hardship. Those are policy choices, not accidents of geography.

4.7 Next steps for this project

A later paper can (i) recover a clean 2016 file, (ii) split results by CMA, (iii) impute liquid assets from the Survey of Financial Security (the V3 model), and (iv) run shock scenarios (rate resets, rent spikes, job loss). Those steps add novelty. They are not required to establish the 15-year residual-income trend.

5 Conclusion

Using 777,707 Census records and a leftover-income test, we find:

1. Housing hardship was 38.0% in 2006 and 37.9% in 2021. Fifteen years of growth did not fix it.
2. Real shelter costs rose 37.1%; real median income rose 33.5%. The 3.6 pp gap is about $456 a year at the 2021 median.
3. The 30% STIR rule improved. Residual-income hardship did not. Those two facts can be true at the same time.
4. Owner hardship rose from 31.4% to 35.6% (2011–2021). Homeownership is not a shield.
5. The 2011–2021 expansion, not the 2008-crisis window, is when leftover income got worse.

The crisis is structural. Another 15 years of the same tools will likely leave about 35–40% of households without enough leftover income after housing.

References

[1] Canada Mortgage and Housing Corporation. *Revised Housing Hardship Methodology: Technical Documentation*. CMHC Research Insights, 2024.

[2] Statistics Canada. *Canadian Housing Survey: Housing Challenges and Experiences, 2024*. Catalogue no. 46-28-0001.

[3] Hulchanski, J. D., Walks, A., and Bourne, L. S. Canada's housing crisis: Dimensions, causes, and policy responses. In T. Bunting and P. Filion (Eds.), *Canadian Cities in Transition* (5th ed., pp. 315–333). Oxford University Press, 2020.

[4] Statistics Canada. *Census 2021 Public Use Microdata File (PUMF): Hierarchical File Documentation*. Catalogue no. 98M0002X, 2022.

[5] Statistics Canada. *Market Basket Measure Thresholds, 2018-base*. Catalogue no. 75F0002M, 2021.

[6] Walks, A., and Twigge-Molecey, A. Income mixing and housing affordability in Canadian cities. *Urban Studies*, 60(11), 2234–2254, 2023.

A Figure files (saved separately)

All figures are 300 dpi PNG files in `papers/paper1_highschool/figures/`. The LaTeX file includes them with `\includegraphics`.

File	Figure	Content
`fig01_national_trends.png`	Fig. 1	Income, shelter, STIR, hardship
`fig02_period_comparison.png`	Fig. 2	2006–2011 vs 2011–2021
`fig03_tenure_divergence.png`	Fig. 3	Owners vs renters
`fig04_income_shelter_squeeze.png`	Fig. 4	Indexed squeeze
`fig05_summary_dashboard.png`	Fig. 5	Four-panel summary

B Table files (saved separately)

Each table exists as CSV (for spreadsheets) and as a `.tex` snippet (for `\input`). Files live in `papers/paper1_highsch`

CSV / TeX stem	Table	Content
`table01_sample_characteristics`	Table 1	Sample and medians
`table02_tenure_hardship`	Table 3	Owner vs renter
`table03_indexed_squeeze`	Table 4	2006 = 100 indexes
`table04_summary_2006_2021`	Table 5	15-year summary
`table05_period_changes`	Table 2	Period changes

To rebuild figures and tables from the JSON results file:

```
python3 papers/paper1_highschool/generate_figures_tables.py
```

To compile this paper:

```
cd papers/paper1_highschool
pdflatex paper1_highschool.tex
pdflatex paper1_highschool.tex
```

C How this version differs from Draft 1.0

Draft 1.0 of the high-school paper used story headings (“What we discovered”) and did not ship separate figure or table files. This version restores a journal skeleton (Abstract, Introduction, Methods, Results, Discussion, Conclusion), saves five figures and five tables as standalone files, and includes them in LaTeX.